Executive Summary - Strategic Heterogeneity of MLB Franchises (2015 - 2024)

MLB franchises do not share a single objective: their observed spending, results and profitability reveal three distinct strategic profiles, two of them stable over time and one cyclical.

Context and research question

Major League Baseball generated a record $12.1 billion in revenue in 2024, yet payrolls range from under $80 million to nearly $400 million. Unlike other North American leagues, MLB has no hard salary cap: spending is moderated only by the Competitive Balance Tax (CBT) and revenue sharing, and 70-80% of revenue is generated locally, so market size matters.

Sports economics has long argued that teams balance win maximization against profit maximization (Sloane, 1971; Fort & Quirk, 1995; Szymanski, 2003). Most empirical work, however, estimates the payroll–performance link while implicitly assuming that all teams pursue the same goal, and rarely tests whether strategies persist over time.

This dissertation asks: **to what extent do the payroll decisions, sporting results and financial outcomes of MLB franchises reveal distinct maximization behaviours over 2015-2024?**

Data and methodology

The study uses a revealed-preference approach (Samuelson, 1948): objectives are inferred from what franchises actually do, not from what they say. The panel covers all 30 franchises over 10 seasons (300 team-seasons), combining payroll, wins, playoff and World Series outcomes, revenue, operating income, margin, CBT paid and market size (sources: Baseball-Reference, Spotrac, Forbes, MLB).

The analysis proceeds in four steps:

1. **Econometrics:** OLS and panel models with franchise and season fixed effects, linking payroll to win percentage.
2. **Efficiency:** cost per win (payroll ÷ wins) as a proxy for how well spending converts into results.
3. **Typology:** k-means clustering (k = 3) on standardized franchise averages of payroll, win %, operating income, margin and cost per win.
4. **Dynamics:** a season-to-season transition matrix and a heatmap of each franchise's group over time.

Key findings

1. Payroll matters, but explains little on its own (H1 confirmed). In OLS, an extra $10M of payroll is associated with about +0.4 percentage points of win percentage, but payroll explains under 10% of the variance ($R^2 = 0.098$). The effect holds with franchise and season fixed effects (coefficients 0.041-0.056, all $p < 0.001$), consistent with diminishing returns: spending raises the odds of winning without guaranteeing it.

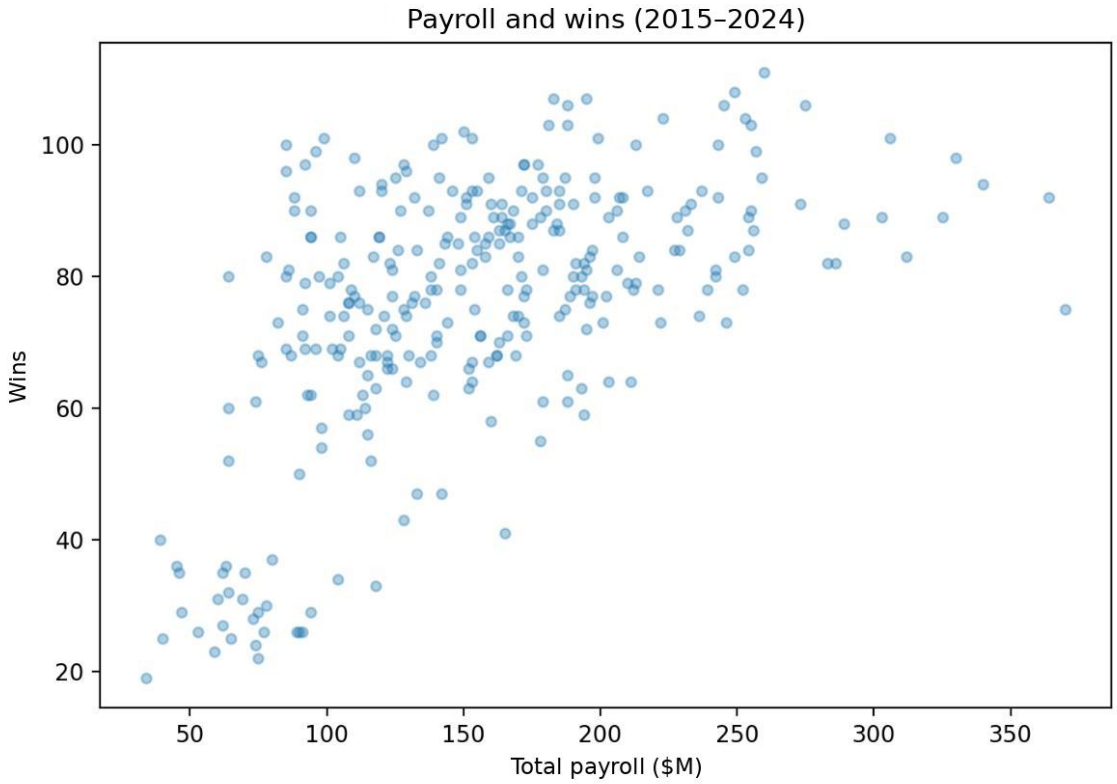


Figure 5 (thesis). Payroll vs wins per team-season, 2015-2024: a positive but widely dispersed relationship.

2. Efficiency differs sharply between franchises (H3 confirmed). The average cost per win is $2.08M, but it ranges from $1.07M for Tampa Bay and $1.27M for Oakland to $2.85M for the New York Yankees. Small-market teams are on average the most efficient, showing that efficiency is a strategic dimension distinct from budget size.

3. Three distinct strategic profiles emerge (H2 and H3 confirmed). The k-means typology goes beyond the classic win-vs-profit dichotomy by revealing a stable intermediate group.

Profile	Franchises	Avg payroll ($M)	Win %	Op. margin	Cost per win ($M)	Big / Mid / Small
G1 · Budget discipline and efficiency	16	125.2	47.9%	+2.5%	1.78	2 / 6 / 8
G2 · Balanced optimization under constraint	11	176.7	51.1%	+6.3%	2.33	9 / 2 / 0
G3 · Win maximization	3	235.4	56.8%	-15.8%	2.79	3 / 0 / 0

Source: thesis Figure 10, franchise averages 2015-2024.

4. Market size shapes capacity, not strategy (H4 partly confirmed). Big-market teams spend more, with a median payroll near $190M versus about $105M for small markets. Yet groups do not map onto market size: some big markets sit in G1 and most G2 members are big markets that stay profitable.

5. Strategies persist, except heavy spending (H5 confirmed). From one season to the next, 78.1% of G1 and 76.2% of G2 observations stay in their group, against only 20% for G3, which mostly returns to G1 (42.9%) or G2 (37.1%). Heavy spending is a temporary "window of contention" rather than a lasting model, though it has become more frequent in recent seasons as league payrolls rise.

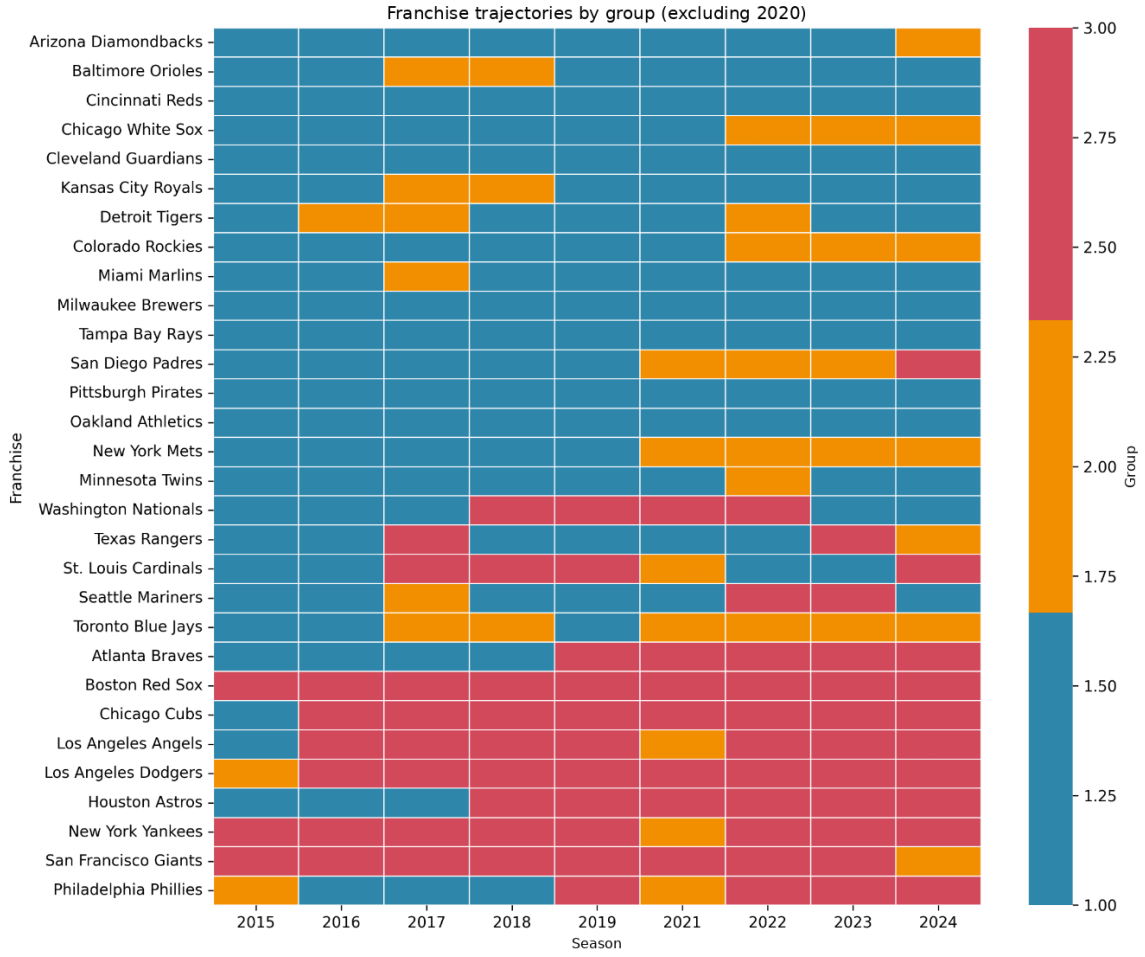


Figure 16 (thesis). Group membership of each franchise by season (blue = G1, orange = G2, red = G3), excluding 2020.

Conclusion and contribution

Strategic diversity in MLB is structural, not residual. The absence of a hard cap, combined with the CBT and revenue sharing, does not push franchises toward one dominant strategy; it frames a space where several stable equilibria coexist, from cost discipline to balanced optimization to cyclical "all-in" spending.

The main contribution is an empirical identification of franchise objective functions from observed behaviour, combining panel econometrics, an efficiency measure, a clustering typology and a dynamic analysis. This moves the literature beyond the payroll–performance elasticity and beyond the binary win-vs-profit framework.

Limitations. Payroll understates true talent investment for teams relying on pre-arbitration players and their farm system, which can inflate apparent efficiency. Results are associations, not strict causal effects, and fixing k = 3 simplifies what may be a continuum of strategies.

Future research. Separating homegrown from acquired talent, and adding player-development proxies such as pre-arbitration WAR, would show whether efficiency stems from contract pricing or from organizational capability.